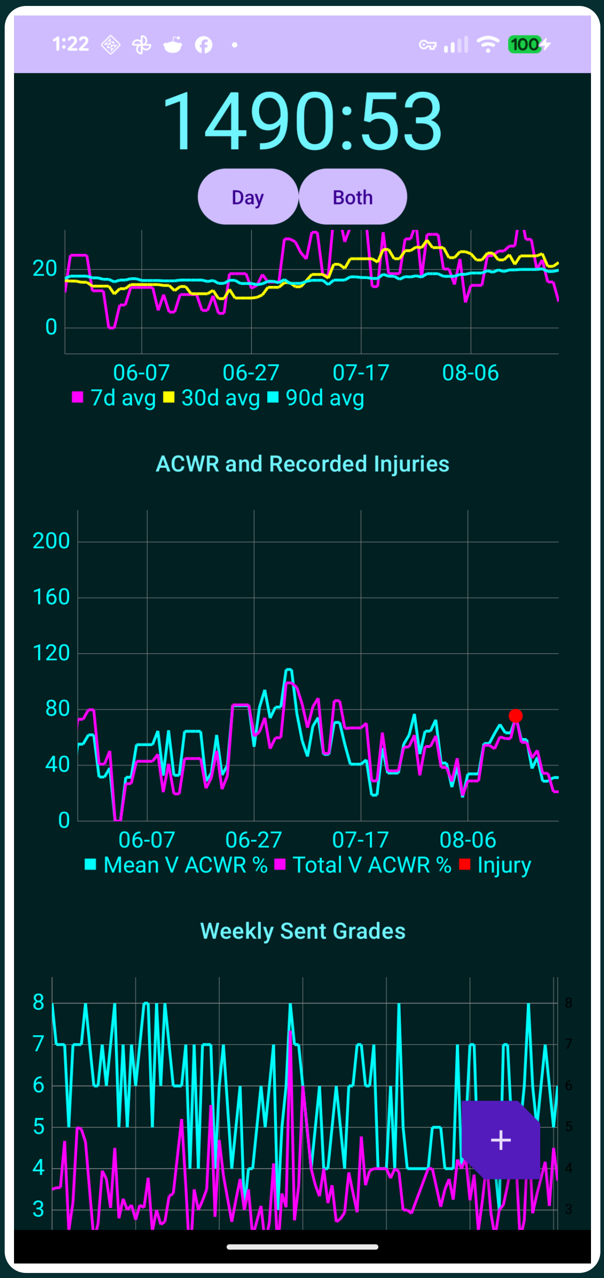
Climbs

User Guide

Record every attempt.
Understand training load.
Keep the data yours.



Android training log and analysis guide

Version: August 2026

1. Start With Consistent Attempts

The app becomes useful when each record means the same thing. The recommended convention is simple: create one climb record for every genuine attempt, including sends and failures.

Recommended definition of an attempt

Count an attempt when you start the climb intending to complete it under your normal send standard. A failure is an attempt that does not meet that standard. The precise boundary is yours, but consistency matters more than choosing a universally perfect rule.

A practical failure rule

- Record a failure when you pull on, meaningfully engage with the sequence, and do not complete the climb.
- A fall, take, dab, skipped move, or incomplete top is a failure if it violates your chosen send standard.
- Beta inspection, brushing holds, or touching moves without a real send intention does not need to count.
- If you record links, rehearsals, or partial attempts, do so every time; otherwise the send-rate models become difficult to compare.

Quick workflow

Metric	Meaning
1. Tap +	Open the entry screen from the home area.
2. Choose type	Select Climb, Hang, or Pull.
3. Enter result	For a climb, enter the V-grade and mark Chuff for failure or Send for success.
4. Add context	Choose Inside or Outside and enter effort, pain, and fear.
5. Save	The current timestamp is supplied automatically and can be edited.
6. Review	Tap an existing record to view it, edit it, or delete it.

Consistency beats precision

Do not reinterpret old failures based on later beta or fitness. Record what happened at the time. When your personal rules change, note the date so trends can be interpreted honestly.

2. Recording Training

Climbs, hangs, and pulls share a timestamp and perceived-effort field, but their training details differ.

Climb fields

Metric	Meaning
Time	ISO-style date and time. It defaults to now; edit it when entering an older session.
Grade	The numeric V-grade. Enter 0 for V0, 5 for V5, and so on.
Chuff	A recorded non-send or failed attempt. Each Chuff contributes an attempted V-grade.
Send	A completed climb under your chosen send standard. A send is also an attempt.
Inside/Outside	Location category used by the global Both/Inside/Outside filter.
Effort	Rating of perceived effort from 1 to 10. Use your subjective effort for that attempt.
Pain	Pain from 0 to 10, default 0. Record the symptom experienced during or immediately after the attempt.
Fear	Fear from 0 to 10, default 0. Use the perceived fear associated with the attempt.

Suggested rating anchors

Scale	Low	Middle	High
Effort 1-10	1-3 easy	4-7 working	8-10 near-max
Pain 0-10	0 none	1-4 noticeable	5-10 substantial
Fear 0-10	0 none	1-5 present	6-10 limiting

Hangs and pulls

- Choose Hang or Pull, then enter weight in pounds and repetitions.
- Effort is available for hangs and pulls as well as climbs.
- Hangs and pulls are stored in the training log but are excluded from climb load and ACWR calculations.

Pain is not a diagnosis

The pain field is a personal log. Stop or modify training when symptoms warrant it and seek qualified medical care for concerning, severe, or persistent symptoms.

3. Navigating the App

Swipe horizontally between four pages. The clock remains fixed at the top and shows elapsed time since the last training record.

Metric	Meaning
Page 1: Log	Main V-points chart, summary metrics, and the chronological list of training records. Tap a row to inspect, edit, or delete it.
Page 2: Events	Record Injury, Bodyweight, and Rating of Perceived Status (RPS). Existing events can also be opened, edited, or deleted.
Page 3: Charts	Scrollable analysis page containing the V-points, moving-average, ACWR, grade, hang/pull, and grade-distribution charts.
Page 4: Settings	Adjust the load baseline and export or back up the database.

Controls shared across pages

Metric	Meaning
Day / Week	Changes compatible time charts between daily and weekly grouping. It does not redefine ACWR, which always uses rolling calendar days.
Both / Inside / Outside	Filters climb records and calculations by location. The visible label is the active filter.
All time / last 3 months	Changes the time scope of the grade-distribution charts on the Charts page.
Baseline: N mo	Sets the prior-history span used by Today and Week load calculations and by the rolling load overlay. Range: 1 to 24 months; current default: 1 month.
Calculate / Update	Runs the more expensive Flash/Redpoint/Project grade progression analysis. The cached result remains until you update it.

Editing and deleting

- Tap a climb, hang, or pull row on Page 1 to open details.
- Use the edit action to change fields and save, or use Delete and confirm.
- Tap an Injury, Bodyweight, or RPS row on Page 2 for the same edit/delete workflow.

Filter awareness

The headline metrics and most charts reflect the current Both/Inside/Outside selection. Check the filter before comparing screenshots or interpreting a sudden change.

4. Headline Metrics

The text below the main chart summarizes the currently selected location. Climb calculations include sends and failures, but not hangs or pulls.

Metric	Meaning
Sends/Day	Average sent V-points across active days in the selected baseline period. A sent V5 contributes 5.
Trys/Day	Average attempted V-points across active days in the selected baseline period. Sends are included because every send is also an attempt.
Load Today	Today's attempted V-points divided by the prior-baseline average for an active climbing day, shown as a percent.
Load Week	Attempted V-points in the current locale calendar week divided by seven expected days at the preceding baseline's per-calendar-day rate.
Load Month	Attempted V-points in the current calendar month divided by attempted V-points in the entire previous calendar month.
ACWR: MV	Rolling Mean V-grade ACWR, normalized to the mean recorded-injury reference value.
ACWR: TV	Rolling Total V-points ACWR, normalized to the mean recorded-injury reference value.
ACWR: Injury	Similarity of the two current ACWR values to the recorded-injury reference center. It is not a calibrated probability of injury.
Flash	Logistic-model estimate of the grade with about a 50% one-attempt send probability, using the last 3 months.
Red	Estimated grade with about a 20.6% per-attempt send probability, equivalent to roughly a 50% chance within 3 independent attempts.
Proj	Estimated grade with about a 5.6% per-attempt send probability, equivalent to roughly a 50% chance within 12 independent attempts.

V-points are grade-weighted volume

Attempting V8 once contributes twice as many V-points as attempting V4 once. These metrics therefore combine attempt count and grade; they do not measure movement count, duration, intensity, or quality directly.

5. Load Calculations

Load uses attempted V-points only. A send is already included as an attempt, so sends are not added a second time.

```
attempted V-points = sum(V-grade for every climb record)

Today % = today's attempted V-points
/ prior baseline average per active day * 100

Week % = current calendar-week attempted V-points
/ (prior baseline average per calendar day * 7) * 100

Month % = current calendar-month attempted V-points
/ previous calendar-month attempted V-points * 100
```

Why Today and Week behave differently

- Today compares with active climbing days only. Rest days are excluded from its baseline denominator.
- Week compares with all calendar days. Rest days are included so a complete seven-day week has a seven-day expectation.
- Week is calendar-based and begins on the first day defined by the phone's locale, typically Sunday in the United States.
- Month is a direct current-month versus previous-month total and does not use the Baseline setting.

Interpreting percentages

Metric	Meaning
100%	Current load exactly matches its comparison baseline.
Below 100%	Current load is lower than the comparison baseline.
Above 100%	Current load is higher. This may be intentional, but inspect the underlying training and recovery context.
0%	There may be no current load, or there may be insufficient non-zero baseline data.

Partial-week effect

Early in a calendar week, the numerator contains only completed days while the denominator still represents seven expected days. The value naturally tends to rise as the week accumulates training.

6. ACWR and Injury Markers

ACWR means acute-to-chronic workload ratio. The app calculates two rolling ratios for every calendar date, using only climb records.

Total V-points ACWR

```
acute total = attempted V-points over trailing 7 calendar days
chronic total = attempted V-points over trailing 28 calendar days
raw Total V ACWR = acute total / chronic total
displayed TV % = raw Total V ACWR / 0.5084 * 100
```

Mean V-grade ACWR

```
daily mean = mean attempted V-grade for that day; rest day = 0
acute mean = mean of daily values over trailing 7 days
chronic mean = mean of daily values over trailing 28 days
raw Mean V ACWR = acute mean / chronic mean
displayed MV % = raw Mean V ACWR / 1.9742 * 100
```

Injury similarity

```
Total Z = (raw TV ACWR - 0.5084) / sqrt(0.0603)
Mean Z = (raw MV ACWR - 1.9742) / sqrt(1.0113)
Injury % = exp[-0.5 * (Total Z^2 + Mean Z^2)] * 100
```

Why ACWR jumps

- The 7-day and 28-day windows have hard edges. A busy day entering the acute window or leaving the chronic window can move the ratio abruptly.
- Rest days lower the 28-day mean, so a concentrated recent week can produce a large Mean V ACWR.
- The normalized Total and Mean percentages naturally top out near 197% and 203%, respectively.
- Values during the first 28 recorded days have incomplete chronic history and should be treated cautiously.

Recorded injuries

Red points on the ACWR chart mark dates entered as Injury events. Their height is placed on the higher ACWR line for visibility; it does not represent injury severity.

Important limitation

The Injury percentage is a mathematical similarity score based on a small set of recorded injury-associated ACWR values. It is not a validated medical risk probability and should not determine whether training is safe.

7. Reading the Charts

Metric	Meaning
V-Points and 7-Day Load	Sent V-points and total attempted V-points over time. On Page 3, rolling 7-day load is overlaid on the right axis for comparison.
V-Points Moving Averages	Past-only averages of daily attempted V-points over 7, 30, and 90 days. Rest days contribute zero. The default view is the latest 3 months; zoom out for older data.
ACWR and Recorded Injuries	Historical Mean V and Total V ACWR percentages with red points for Injury events. The default view is 90 days and can be zoomed out.
Weekly Sent Grades	Maximum sent grade and average sent grade for each seven-day block containing sends.
Flash / Redpoint / Project	Rolling 90-day logistic estimates of one-attempt, 3-attempt, and project-range grades. Press Calculate or Update to refresh.
Hang and Pull Volume	Training-volume view based on recorded weight and repetitions, grouped by the Day/Week control.
Send Probability by Grade	Observed send fraction at each grade. Use the all-time/last-3-months control to change scope.
V-Points Sent by Grade	Distribution of sent V-points across grades for the selected time scope.
Attempts by Grade	Distribution of recorded attempts across grades for the selected time scope.

Chart interaction

- Drag horizontally to inspect older or newer dates.
- Pinch to zoom. Time charts preserve older history even when they open to a recent default window.
- Use Day/Week for broad trend inspection and Both/Inside/Outside to compare environments.
- Sparse data can make fitted Flash/Red/Proj estimates unstable or unavailable.

Correlation is context, not causation

Overlaying load, ACWR, grades, and injury dates can reveal patterns worth investigating. It cannot establish that one metric caused an injury or performance change.

8. Events, Export, and Backup

Events provide context that is separate from individual climb attempts. Swipe to Page 2 to add them.

Metric	Meaning
Injury	Timestamp plus a free-text note. Injury dates appear as red markers on the ACWR chart.
Bodyweight	Timestamp plus a floating-point bodyweight value. Keep units consistent across entries.
RPS	Rating of Perceived Status from 0 to 10. Choose personal anchors and keep them stable; for example, 0 very poor and 10 excellent.

CSV export

- On Page 4, tap Export. The app writes climb_data.csv to the phone's Downloads collection.
- The CSV contains climb/hang/pull rows followed by Injury, Bodyweight, and RPS events.
- CSV is the easiest format to inspect, archive, or import into a spreadsheet.

Database backup

- Tap Backup to write my_database_backup.db to Downloads.
- Move the file off the phone using Files, USB transfer, cloud storage, or Android Studio Device Explorer.
- Keep dated copies instead of repeatedly trusting a single file.

Recommended backup routine

After important sessions or before installing a new build: export CSV, create a database backup, move both files off the phone, and verify that their file sizes are non-zero. The CSV is the portable safety copy; database restoration is a developer workflow.

Database caution

Android Room may use write-ahead logging. Because the current Backup action copies the main database file, critical archives should always include the CSV export too. A future backup implementation can checkpoint or package the database with its WAL state.

9. Install From GitHub

The easiest installation method is to download the APK directly on the Android phone. A computer and Developer Options are not required.
Download app-release.apk

A. Install directly on the phone

- Open the GitHub repository on the phone and download app/release/app-release.apk, or tap the download link above.
- If GitHub opens a file page, tap the download button or View raw.
- Open the downloaded APK from the browser notification or Downloads folder.
- If Android blocks it, follow the prompt to let the browser or Files app Install unknown apps.
- Return to the installer, tap Install, then open Climbs from the app drawer.

Updates and security

- Android warns because this APK comes from outside the Play Store. Only install the APK from this repository.
- Export CSV and create a database backup before updating.
- Install the new APK over the existing app. Do not uninstall first unless you intend to remove local app data.
- The Install unknown apps permission can be disabled again after installation.

B. Developer installation

Android Studio and ADB are optional paths for developing or testing a local source build.

- Enable Developer Options by tapping Build number seven times in About phone, then enable USB debugging.
- Clone the repository, open it in Android Studio, connect the authorized phone, and press Run.

```
./gradlew :app:assembleDebug
adb devices
adb install -r app/build/outputs/apk/debug/app-debug.apk
```

If installation is blocked

Confirm the APK finished downloading, allow installation from the app that opened it, and make sure the device has enough free storage. A signature mismatch may require the same signing certificate used by the existing installation.

10. Good Data Habits

A useful training log does not need perfect physiology. It needs stable definitions and enough context to make comparisons fair.

Metric	Meaning
Record promptly	Log each attempt near the time it happened so timestamps and subjective ratings remain credible.
Use one send rule	Decide how you handle dabs, takes, starts, and tops, then apply the rule consistently.
Record failures	Leaving out failures inflates send probability, grade estimates, and apparent performance.
Keep grades stable	Use the grade presented or a consistent personal grade system; avoid changing old entries to match later opinions.
Use location	Inside and outside grades differ. Correct location labels make the filter meaningful.
Anchor ratings	Define what effort, pain, fear, and RPS numbers mean to you and avoid changing anchors frequently.
Add injury notes	Describe area, onset, and context in neutral language. Do not use the app as a substitute for assessment.
Back up often	Keep both CSV and database copies outside the phone, especially before app updates or device changes.

A note on interpretation

Metrics are summaries of what was recorded. Missing failures, inconsistent grades, travel, gym-setting changes, and selective logging can all produce apparent trends. Use charts to ask better questions, then compare them with sleep, stress, recovery, and the actual texture of training.

> **Short version**
>
> Record every genuine attempt. Mark sends honestly. Keep your definition of failure stable. Use load and ACWR as context, not commands. Export before updates. Back up somewhere that is not the phone.

Source code and updates:
https://github.com/graysomb/basic-android-kotlin-compose-training-inventory-app-main
This guide describes the application behavior present in the repository in August 2026.